

Toward Scalable Multicore Fault-Tolerant Quantum Computers using Quantum Multiplexing

Shin Nishio

*Graduate School of Science and Technology, Keio University, Japan
Department of Physics and Astronomy, University College London, UK*

Biography

Shin Nishio specializes in system software for quantum computing and quantum error correction codes. He started working in quantum information processing when he was in the undergraduate program under the supervision of Prof. Rodney Van Meter at Keio University. As an undergraduate, he worked with Prof. Takahiko Satoh and Dr. Atsushi Matsuo to design problems and judge submissions for the IBM Quantum Challenge in 2019, the first quantum programming competition hosted by IBM. Over 700 people participated in the challenge and enjoyed designing quantum algorithms for a toy graph theory problem. He earned his Ph.D. in informatics at SOKENDAI in alliance with National Institute of Informatics and Okinawa Institute of Science and Technology under the supervision of Prof. Kae Nemoto, where he conducted research on reduction of resource overheads for distributed fault-tolerant quantum computing. Currently Shin is a project assistant professor in Prof. Takahiko Satoh's Group at Keio University in Japan. He is also a research associate funded by Overseas Research Fellowships of JSPS in Prof. Dan Browne's Group at University College London in UK. He leads efforts to define the functions required for large-scale quantum computing systems and to address the challenges involved in their realization.

Abstract

Linking multiple quantum processors via a quantum interconnect is a promising approach to realizing large-scale, fault-tolerant quantum computation. Quantum multiplexing refers to encoding high-dimensional quantum information onto a single photon by exploiting multiple degrees of freedom or multiple components within a single degree of freedom. This technique is expected to enable more efficient quantum communication.

In this work, we consider the adoption of quantum multiplexing in an interconnected fault-tolerant quantum computing architecture. We analyze how such multiplexing affects the logical error rate of an error-corrected communication channel using surface codes and hypergraph product codes. While multiplexing can offer significant resource savings, it also converts photon-loss errors into correlated errors, which can degrade code performance. However, we demonstrate that this detrimental effect can be substantially mitigated by employing an appropriate strategy for assigning qubits to photons—making multiplexed communication viable with both higher throughput and improved fidelity. We also place our analysis in the context of recent developments in fault-tolerant quantum computation, highlighting how quantum multiplexing aligns with emerging architectural trends.